Name

Mathematics Stage 5 Core

Surface Area of Prisms and Cylinders

Book 1

Version: 261006

Feedback:
https://MrDingMaths.com

Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-ARE-C-01 solves problems involving the surface area of right prisms and practical problems involving the area of composite shapes and solids

Solve problems involving areas and surface areas

- Solve practical problems involving the areas of composite shapes
- Identify the edge lengths and the faces making up the surface area of prisms
- Recognise and justify whether a diagram represents a net of a right prism
- Create and rearrange nets of right prisms
- Find the surface areas of prisms, given their nets, excluding curved surfaces
- Solve problems involving surface areas of prisms, excluding curved surfaces

Develop and apply the formula for surface areas of cylinders

- Recognise the curved surface of a cylinder as a rectangle and apply this knowledge to calculate the area of the curved surface
- Develop and apply the formula to find the surface area of a closed cylinder: $A = 2\pi r^2 + 2\pi rh$, where r is the length of the radius and h is the perpendicular height

Solve problems involving surface areas of cylinders and related composite solids

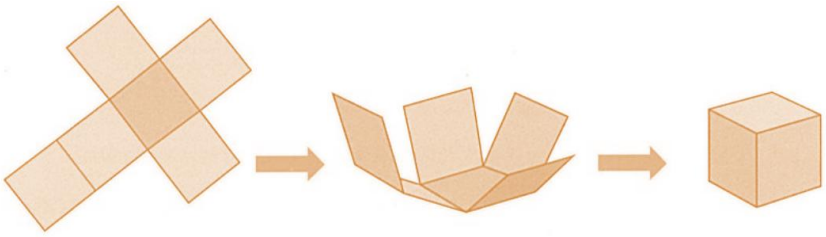
- Solve problems involving surface areas of cylinders and related composite solids

Nets of 3D solids

ⓘ Nets

A **net** shows a 3D shape's faces laid flat.

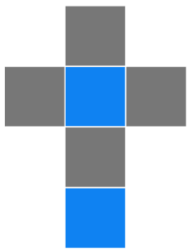
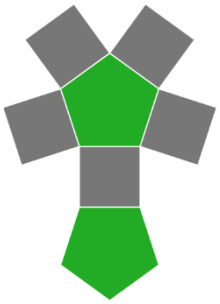
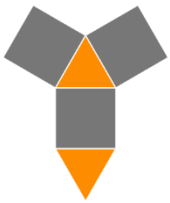
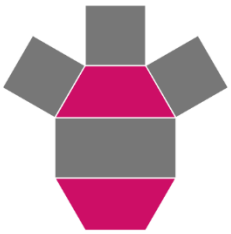
A **prism** has two identical faces connected by rectangles.



Identify nets of right prisms

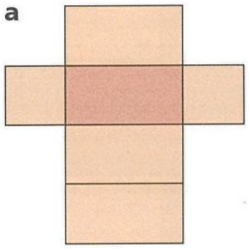
Name the prism that is made from each net.

https://mathigon.org/polypad/Qrx1t9V9RTSqA

a 	**b** 	**c** 	**d** 
e 	**f** 	**g** 	**h** 

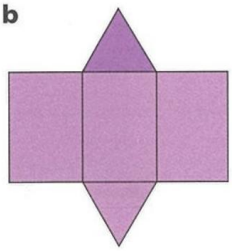
1 Give the name of each of the 3-D shapes that can be made from these nets.

a



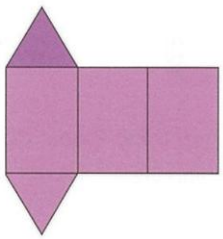
Rectangular prism

b



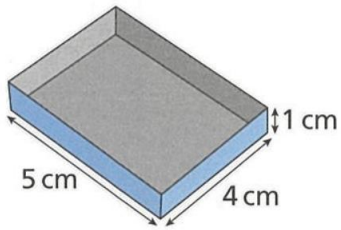
Trianglular prism

c



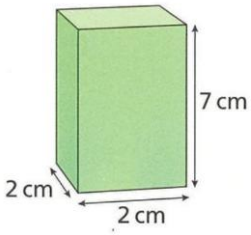
Triangular prism

2 Sketch an accurate net for this open box.

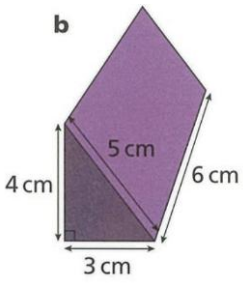


3 Sketch a net for each of these shapes.

a

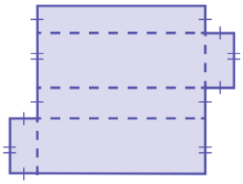


b



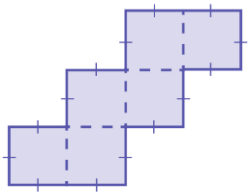
4 Name the solid made by the net.

a



Rectangular prism

b



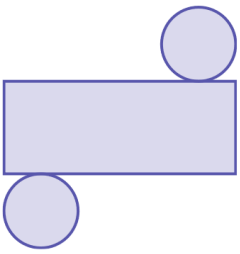
Cube

c



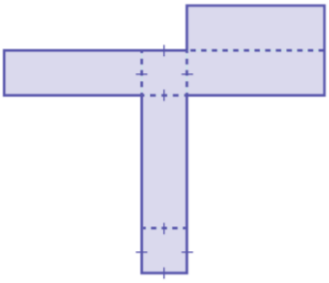
Triangular prism

d



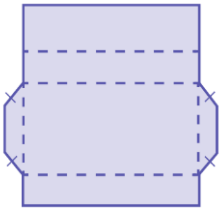
Cylinder

e



Square prism

f



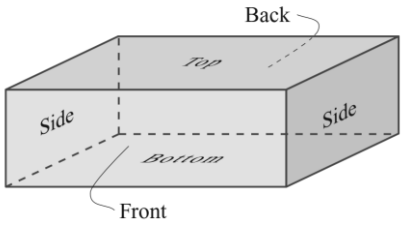
Trapezoidal prism

Surface Area

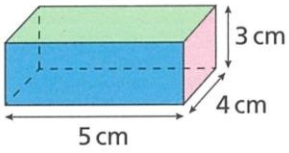
Surface Area

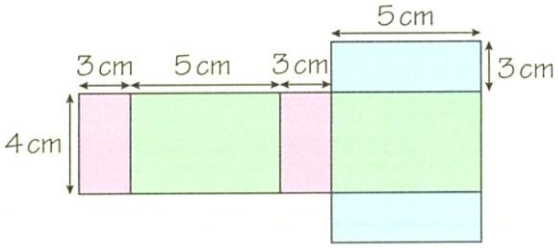
The **surface area** of a prism is the total area of all its faces.

It equals the area of the prism's **net**.



Example Calculate surface area of prisms given their nets





Area of front and back: $(5 \times 3) \times 2 = 30$

Area of sides: $(4 \times 3) \times 2 = 24$

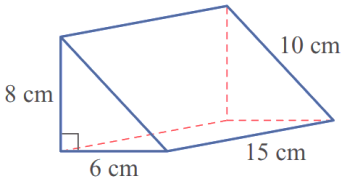
Area of bottom and top: $(5 \times 4) \times 2 = 40$

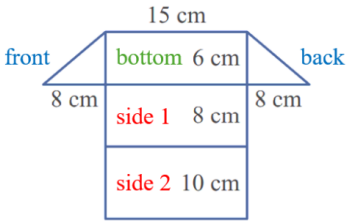
Total surface area: $30 + 24 + 40$

$= 94 \text{ cm}^2$

Guided Practice

Calculate the surface area





Area of front and back:

Area of side 1:

Area of side 2:

Area of bottom:

Total surface area:

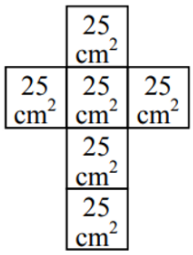
408 cm^2

Key Ideas

1. The …… …… of a prism is the area of the outer parts of the solid.
   It can be calculated by adding up the areas of all of the ……

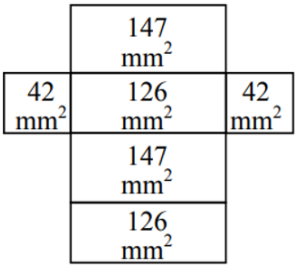
1 Find the surface area.

a



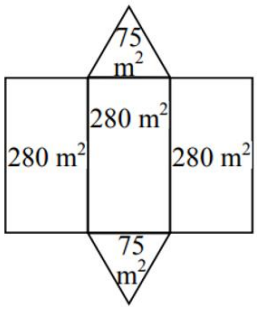
150 cm²

b



630 mm²

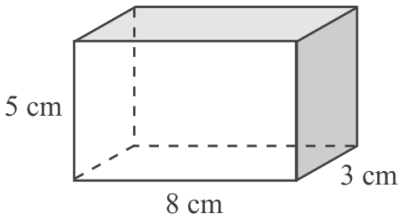
c

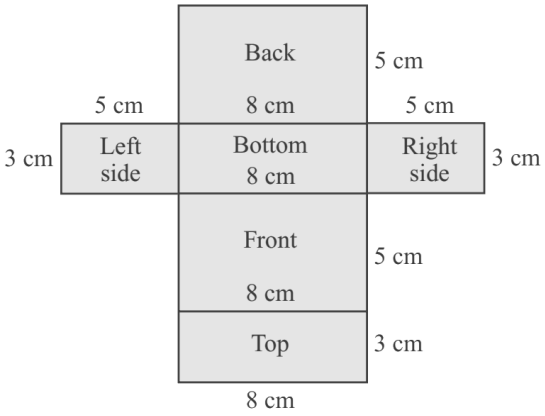


990 m²

2 Calculate the surface area of these prisms.

a





Area of front and back: ..... (5 × 8) × 2

Area of sides: ..... (3 × 5) × 2

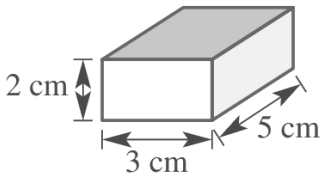
Area of bottom and top: ..... (3 × 8) × 2

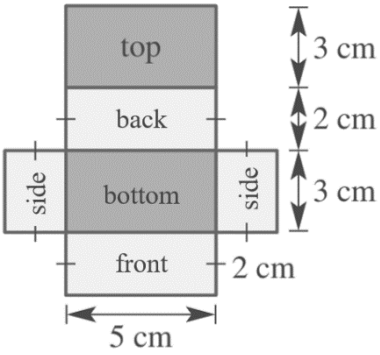
Total surface area: .....

= .....

158 cm²

b





Area of front and back: ..... (3 × 2) × 2

Area of sides: ..... (5 × 2) × 2

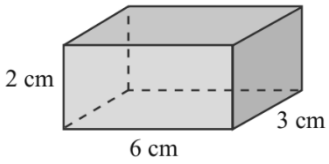
Area of bottom and top: ..... (3 × 5) × 2

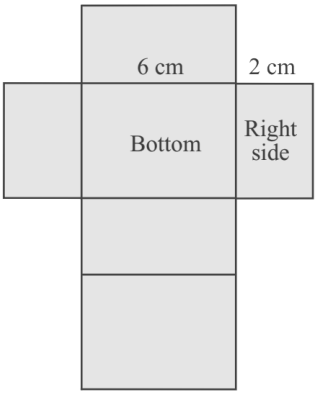
Total surface area: .....

= .....

62 cm²

c





Area of front and back: ..... $(6 \times 2) \times 2$

Area of sides: ..... $(3 \times 2) \times 2$

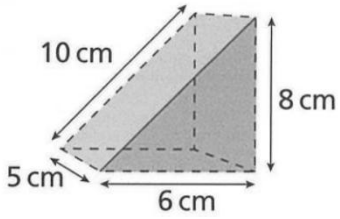
Area of bottom and top: ..... $(6 \times 3) \times 2$

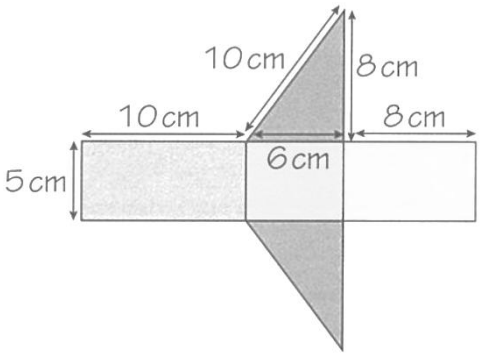
Total surface area: .....

= .....

72 cm²

d





Area of front and back: ..... $\left(\frac{6 \times 2}{2}\right) \times 2$

Area of left side: ..... 5×10

Area of right side: ..... 5×8

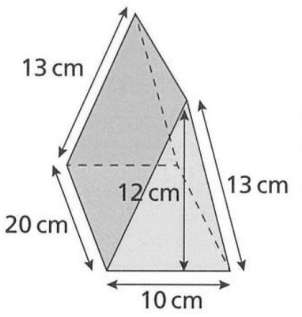
Area of bottom: ..... 6×5

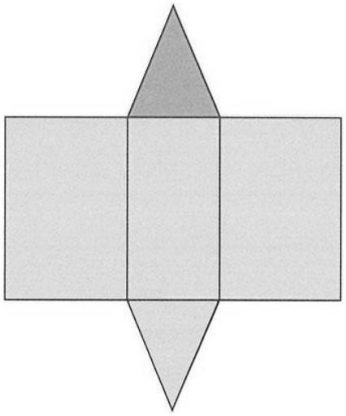
Total surface area: .....

= .....

132 cm²

e





Area of front and back: ..... $\left(\frac{10 \times 12}{2}\right) \times 2$

Area of left side: ..... 20×13

Area of right side: ..... 20×13

Area of bottom: ..... 10×20

Total surface area: .....

= .....

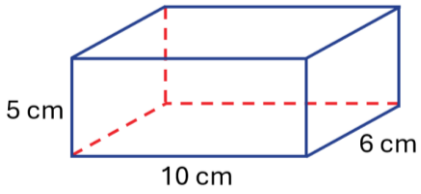
840 cm²

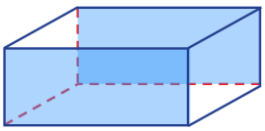
Surface Area of a Rectangular Prism without a Net

Surface Area

To find **surface area**, picture the faces covering the 3D shape.
This avoids having to draw out the net.

Example Calculate surface area of a rectangular prism without a net

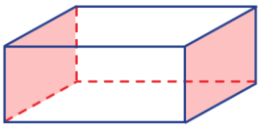




front and back rectangles

$(10 \times 5) \times 2$

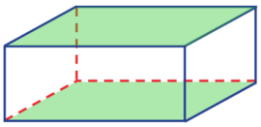
$= 100$



side rectangles

$(5 \times 6) \times 2$

$= 60$



top and bottom rectangles

$(10 \times 6) \times 2$

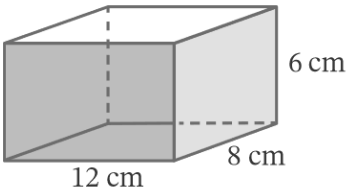
$= 120$

Total surface area $= 100 + 60 + 120$

$= 280 \text{ cm}^2$

Guided Practice

Calculate the surface area of this rectangular prism.



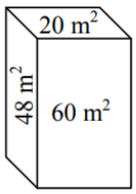
front and back	side	top and bottom

Total surface area:

432 cm²

1 Find the surface area of these solids.

a



Area of front and back

60 × 2

Area of sides

48 × 2

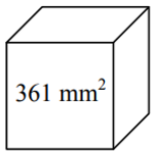
Area of top and bottom

20 × 2

Total surface area

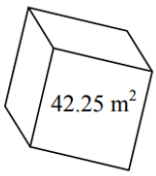
256 m²

b



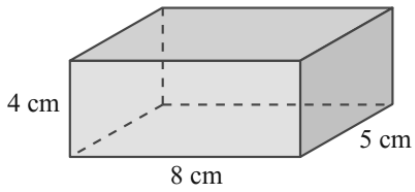
2166 m²

c



253.5 m²

2 Complete the following the calculate the surface area of the rectangular prism.



Area of front and back = (8 × ……) × 2 = ……cm²

(8 × 4) × 2

Area of both sides = (…… × ……) × 2 = …… cm²

(5 × 4) × 2

Area of top and bottom = (…… × ……) × 2 = …… cm²

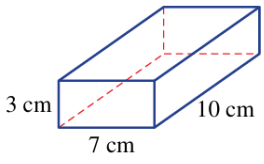
(8 × 5) × 2

Total surface area = …… + …… + …… = …… cm²

184 cm²

3 Find the surface area of the following prisms.

a



Front and back:

(7 × 3) × 2

Sides:

(3 × 10) × 2

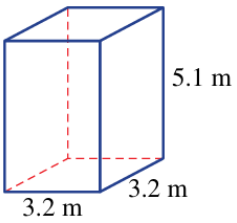
Top and bottom:

(7 × 10) × 2

Total surface area

242 cm²

b



Front and back:

(3.2 × 5.1) × 2

Sides:

(3.2 × 5.1) × 2

Top and bottom:

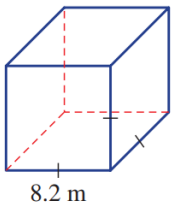
(3.2 × 3.2) × 2

Total surface area:

85.76 m²

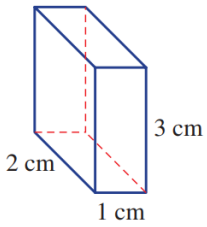
4 Find the surface area of the following prisms.

a



403.4 cm²

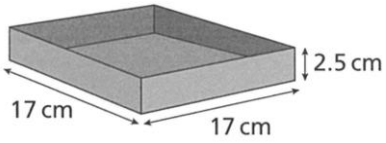
b



22 cm²

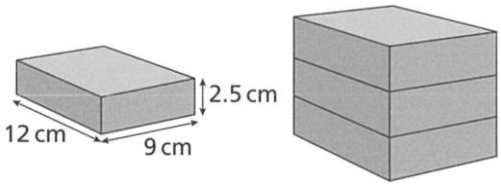
5 Beca makes an open box. She is going to paint the outside of the box.

Work out the total area to be painted.



459 cm²

6 Ed stacks three identical rectangular prisms on top of each other as shown.



He says "The surface area of my shape will be three times as large as the surface area of one of the rectangular prisms."

Explain why he is wrong.

When you stack the prisms, two faces are no longer on the surface, so the surface area will be less.

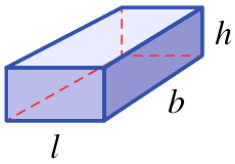
7 A book measuring 20 cm long, 15 cm wide and 3 cm thick is covered in plastic.

What area of plastic is needed to cover 1000 books? Convert your answer to m^2.

SA of 1 book: 810 cm²

SA of 1000 books: 810 000 cm² ÷ 100² = 81 m²

8 Consider the rectangular prism shown.



a Write an expression for the surface area of the prism.

$SA = 2lb + 2lh + 2bh$

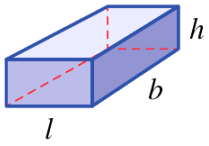
b By factorising, show that the formula for the surface area for any rectangular prism is $SA = 2(lb + lh + bh)$.

Factorise out common factor of 2 from expression in part a.

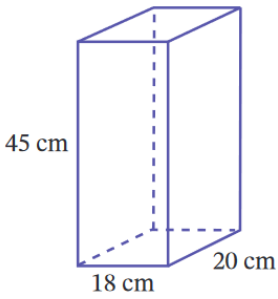
9 Use this formula to quickly calculate the surface area of these rectangular prisms.

ⓘ Surface Area of a Rectangular Prism

$SA = 2(lb + lh + bh)$

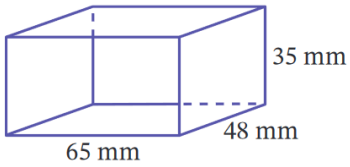


a



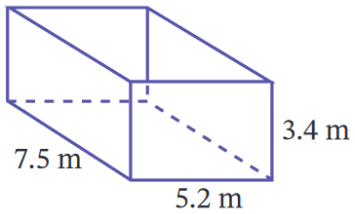
4140 cm²

b



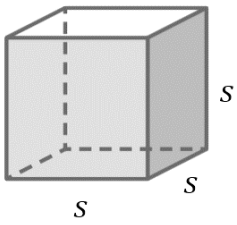
14 150 mm²

c



164.36 m²

10 What would be the formula for the surface area of a cube with side length s?



$6s^2$

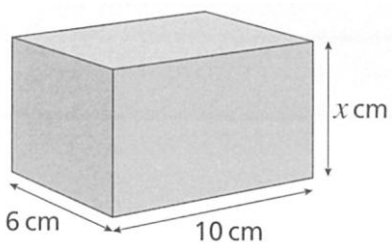
11 A rectangular prism storage crate measures 0.8 m by 1 m by 1.5 m.
Claudia paints all the faces of the crate.
One tin of paint costs \$8. Each tin covers 4 m^2
Work out the total cost of the paint that Claudia needs.

Surface area = 7 m²

2 paint tins needed

$\$8 \times 2 = \$1$

12 The surface area of this rectangular prism is 376 cm^2. Work out the value of x.



$376 = 2(60 + 6x + 10x)$

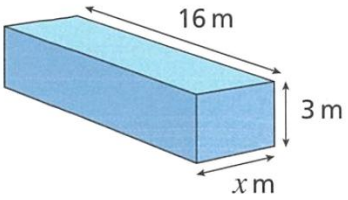
$188 = 60 + 16x$

$x = 8$ cm

Work out the value of x in each of these rectangular prisms.

a

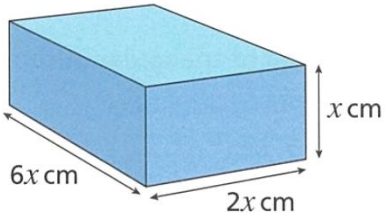
surface area = 248 m^2



4 m

b

surface area = 90 cm^2



1.5 cm

13 A cube has surface area 150 m^2. Determine the length of each side of the cube.

$150 = 6x^2$

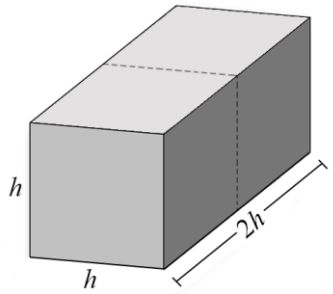
$x^2 = 25$

$x = 5$ m

14 NAPLAN A+

The rectangular prism, shown below, is cut in half to form 2 equal cubes.

What is the ratio of the surface area of the rectangular prism to one of the cubes?



SA of 1 cube = $6h^2$

SA of prism = $2(h^2 + 2h^2 + 2h^2) = 10h^2$

Ratio of prism to cube = $10h^2 : 6h^2$

$= 5:3$

15 A rectangular prism has dimensions in the ratio 1:3:5. It has a volume of 120 cm^3. Find:

a the dimensions of the prism.

Let sides be $x, 3x, 5x$

$V = Ah$

$120 = 15x^3$

$x = 2$

Dimensions are 2 cm, 6 cm, 10 cm

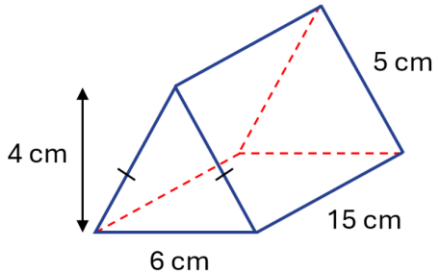
b the surface area of the prism.

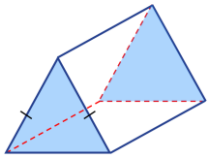
$SA = 2(2 \times 6 + 2 \times 10 + 6 \times 10)$

$= 184$ cm²

Surface Area of a Triangular Prism without a Net

Example Calculate surface area of a triangular prism without a net

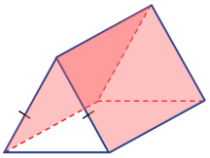




front and back triangles

$\left(\frac{6\times4}{2}\right)\times 2$

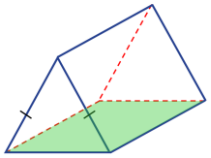
$= 24$



side rectangles

$(15\times 5)\times 2$

$= 150$



bottom rectangle

6×15

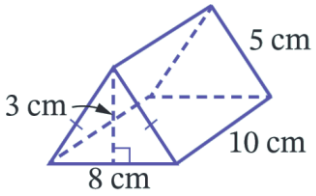
$= 90$

Total surface area $= 24 + 150 + 90$

$= 264\ \text{cm}^2$

Guided Practice

Calculate the surface area of this triangular prism.



front and back triangles

side rectangles

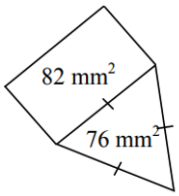
bottom rectangle

Total surface area:

204 cm²

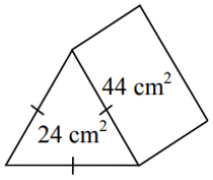
1 Calculate the surface area of these triangular prisms.

a



398 mm²

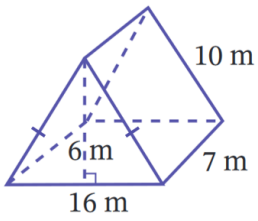
b



180 cm²

2 Calculate the surface area of these triangular prisms.

a



Area of front and back: ..... $\left(\frac{16 \times 6}{2}\right) \times 2$

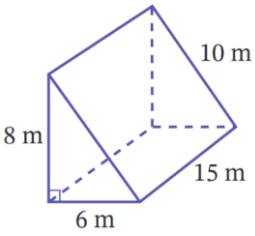
Area of sides: ..... 7×10

Area of bottom: ..... 7×16

Total surface area: .....

348 cm²

b



Area of front and back: ..... $\left(\frac{6 \times 8}{2}\right) \times 2$

Area of left side: ..... 15×8

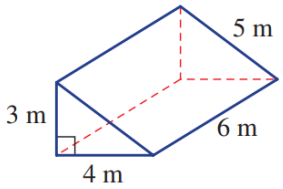
Area of right side: ..... 15×10

Area of bottom: ..... 15×6

Total surface area: .....

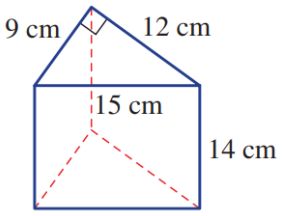
408 cm²

c



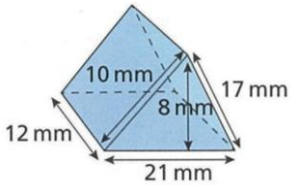
84 m²

d



612 cm²

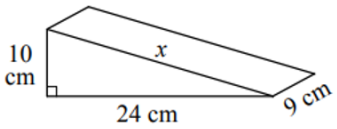
3 Find the surface area of the triangular prism.



744 mm²

4 Calculate the missing length using Pythagoras' Theorem, then calculate the surface area.

a



$x^2 = \square^2 + \square^2$

$x^2 = \square$

$x = \sqrt{\square}$

$x = \square$ cm

$SA = 2 \times ½ \times \square \times \square$

$+ \square \times 9$

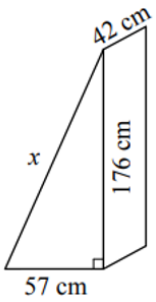
$+ \square \times 9$

$+ \square \times 9$

$SA = \square$ cm^2

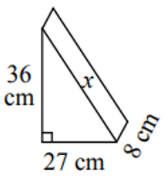
$x = 26$ cm, $SA = 780$ cm²

b



$x = 185$ cm, $SA = 27\,588$ cm²

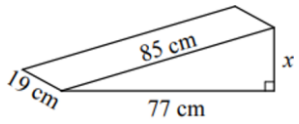
c



$x = 45$ cm, $SA = 1836$ cm²

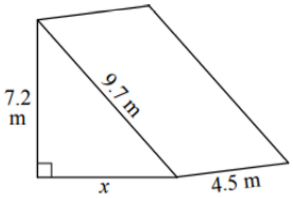
For these triangles, you are finding a shorter side, not the hypotenuse.
Calculate the surface area.

d



$x = 36$ cm, $SA = 6534$ cm²

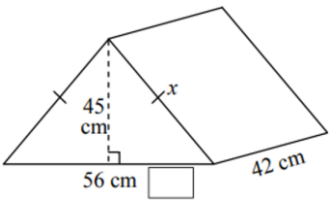
e



$x = 6.5$ cm, $SA = 152.1$ m²

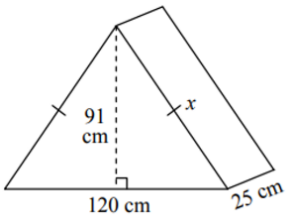
Halve the base, then find the missing length.
Calculate the surface area.

f



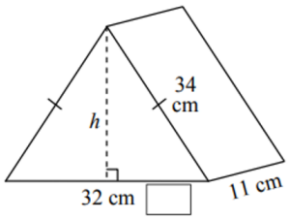
$x = 53$ cm, $SA = 9324$ cm²

g



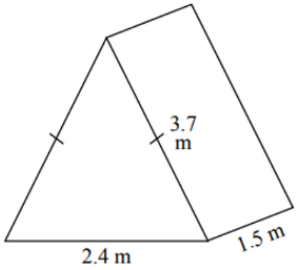
$x = 109$ cm, $SA = 19\,370$ cm²

h



$h = 30$ cm, $SA = 2060$ cm²

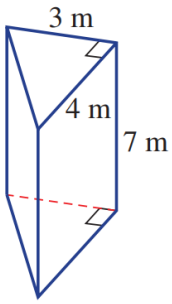
i



$h = 3.5$ m, $SA = 23.1$ m²

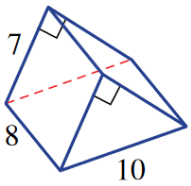
5 Use Pythagoras' Theorem $h^2 = a^2 + b^2$ to find unknown lengths and hence calculate the surface area of these solids. Round to 1 d.p. where necessary.

a



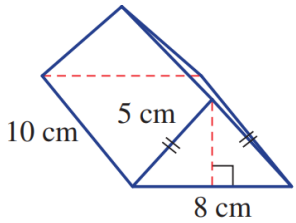
96 m²

b All units in cm.



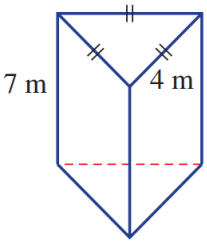
243.1 cm²

c



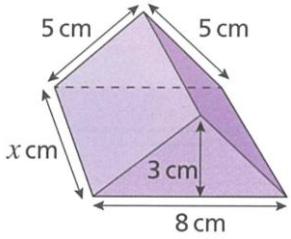
204 cm²

d



97.9 m²

6 This prism has surface area 222 cm². Determine the value of x.



$\left(\frac{3 \times 8}{2}\right) \times 2 + 5x + 5x + 8x = 222$

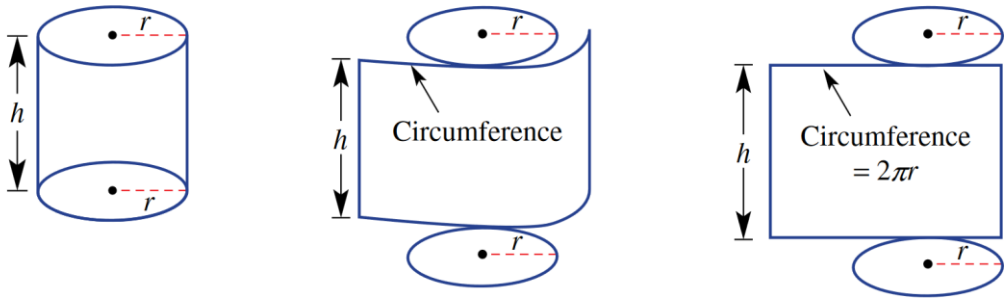
$18x = 198$

$x = 11$ cm

Surface Area of Closed Cylinders

ⓘ Net of a Cylinder

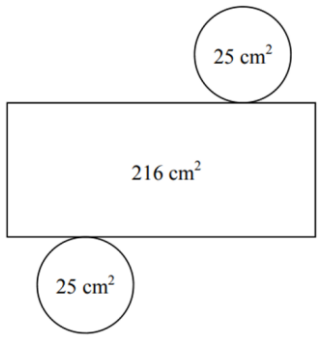
A **cylinder's net** has two circles and one rectangle.



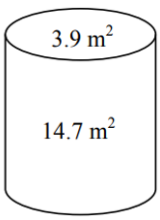
🔍 Surface Area of a Cylinder

Calculate the surface area of these cylinders by adding the areas of each face.

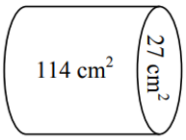
a



b



c

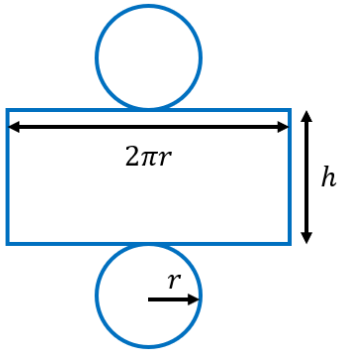


A cylinder is made up of ..... faces.

The top and bottom are circles, with area .....

The curved part can be unrolled into a rectangle,

with length ..... and width .....



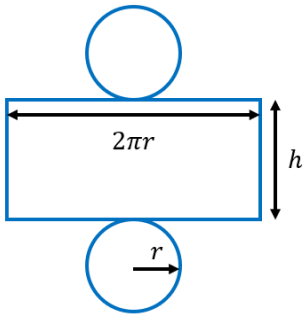
So the total surface area is A = ... + ...

Surface Area of a Closed Cylinder

$$SA = 2\pi r^2 + 2\pi rh$$

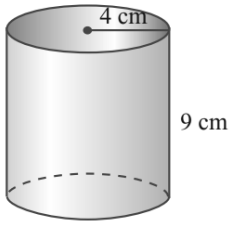
$2\pi r^2$: 2 circles

$2\pi rh$: curved surface



Example Calculate surface area of a cylinder

Calculate the surface area of the cylinder, rounded to 2 decimal places.



$$A = 2\pi r^2 + 2\pi rh$$
$$= 2\pi(4)^2 + 2\pi(4)(9)$$
$$= 326.73 \text{ cm}^2$$

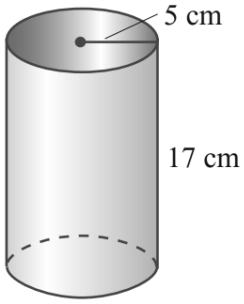
$2\pi r^2$ represents the .....

$2\pi rh$ represents the .....

Guided Practice

Calculate the surface area of these cylinders.

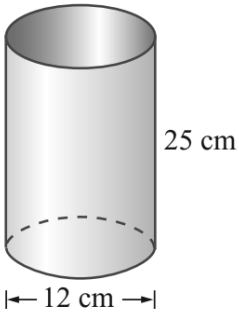
a



$$A = 2\pi r^2 + 2\pi rh$$
$$=$$
$$=$$

691.15 cm²

b



1168.67 cm²

Key Ideas

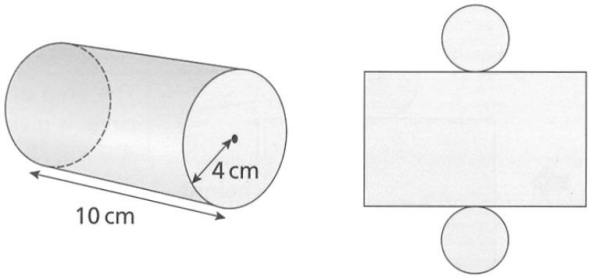
1. A cylinder is made up of two ..... and a single .....
2. The rectangle's dimensions are the height of the cylinder and the ..... of the circles that make up the bottom and top.

1 Consider the cylinder and net to the right.

a Label the net with correct lengths.

b Calculate the area of:

i The two circles.
Answer in exact form.



$2\pi(4)^2 = 32\pi$

ii The rectangle. Answer in exact form.

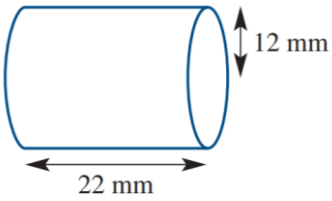
$2\pi(4)(10) = 80\pi$

iii Hence find the surface area of the cylinder to 1 d.p.

351.9 cm²

2 Find the surface area of these cylinders. Round your answer to 1 decimal place.

a



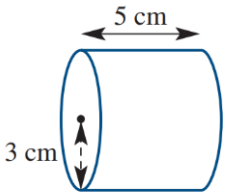
$A = 2\pi r^2 + 2\pi rh$

$= \ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots$

2563.5 mm²

b



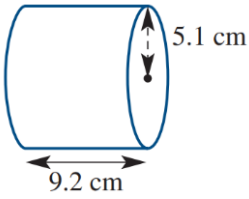
$A = \ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots$

150.8 cm²

c



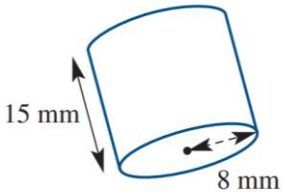
$A = \ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots$

$= \ldots\ldots\ldots\ldots$

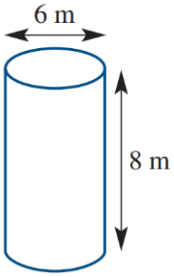
458.2 cm²

d



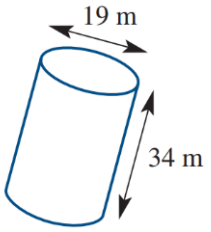
1156.1 mm²

e



207.3 cm²

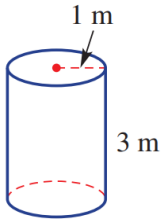
f



2596.5 cm²

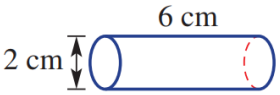
3 Work out the surface area of each cylinder. Answer in exact from in terms of π.

a



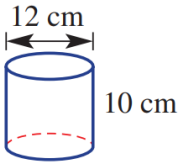
8π m²

b



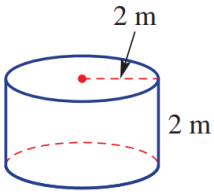
14π cm²

c



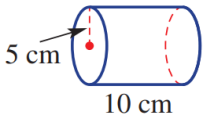
192π cm²

d



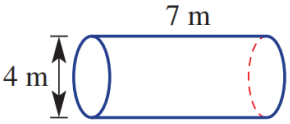
16π m²

e



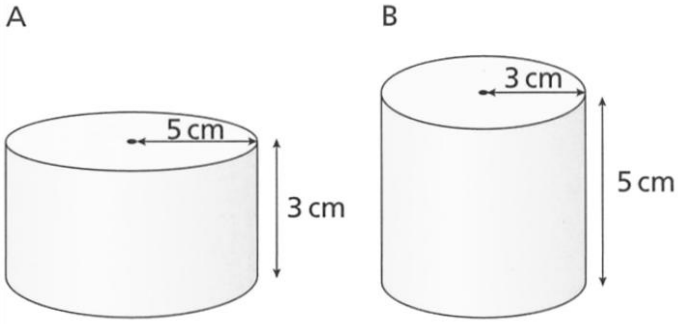
150π cm²

f



36π m²

4 Which shape has the larger surface area? Justify your answer.



A: 80π cm²

B: 48π cm²

A has larger surface area.

5 A cylinder has a radius of 12 cm and a height of 20 cm.

a Determine the surface area of the cylinder to 1 d.p.

2412.7 cm²

b The height of the cylinder is increased by 10%.

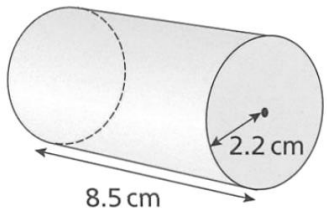
Work out the new total surface area of the cylinder.

2563.5 cm²

c Work out the percentage increase in the total surface area.

6.25% increase

6 Express the curved surface area of this cylinder as a percentage of the total surface area. Give your answer to 2 decimal places.



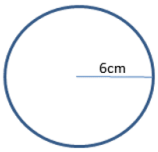
Curved: $2\pi rh = 117.5$ cm²

Total = 147.9 cm²

Percentage = 79.44%

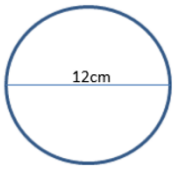
7

a The radius of a circle is 6 cm. Calculate the area of the circle.
Leave your answer in terms of π



36π cm²

b Shown is a face of a cylinder with a height of 10 cm. Calculate the exact surface area of the cylinder.



192π cm²

c The circumference of a circle shown is 7π cm. Calculate the area of the shaded region in terms of π.



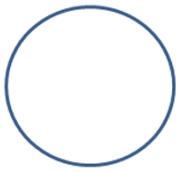
$C = 2\pi r$

$7\pi = 2\pi r$

$r = 3.5$ cm

$A = \frac{\pi(3.5)^2}{4} = \frac{49}{16}\pi$ cm²

d The area of the circle is 16π cm². Find the circumference to 2 decimal places.



$A = \pi r^2$

$16\pi = \pi r^2$

$r = 4$ cm

$C = 2\pi r = 8\pi$ cm

8 A cylinder has radius 6 cm and height of h. The curved surface area of the cylinder is 754 cm²

Work out the value of h.

$2\pi(6)h = 754$

$h = 20$ cm

9 A cylinder has radius 4 m and height h. The total surface area of the cylinder is 128π m².

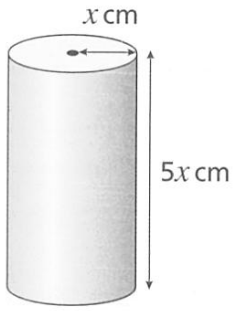
Work out the value of h.

$2\pi(4)^2 + 2\pi(4)(h) = 128\pi$

$8\pi h = 96\pi$

$h = 12$ m

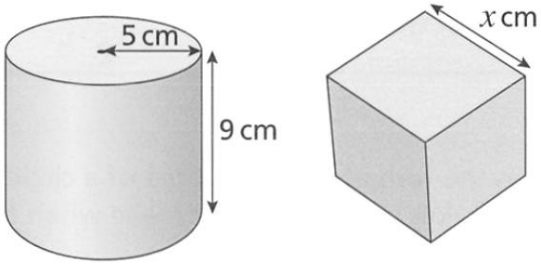
10 This cylinder has a surface area of 972π cm^2. Work out the value of x.



$972\pi = 2\pi x^2 + 2\pi x(5x)$

$x = 9$

11 The cylinder and cube have the same surface area. Work out the value of x to 2 d.p.



SA of cylinder = 140π cm²

$6x^2 = 140\pi$ cm²

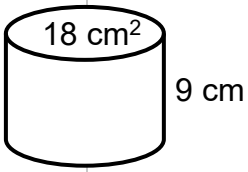
$x = 8.56$ cm

12 If you double the height of a cylinder, do you double its surface area?

Explain why.

No. Only the curved surface is doubled, and the circular ends stay the same, so the total is not doubled.

13 Consider the cylinder below.



a Find the diameter of the cylinder to 2 d.p.

4.79 cm

b Calculate the volume of the cylinder.

162 cm³

c The top surface of a cylinder with the same volume has area of 50 cm^2. What is the height of the cylinder?

$V = \pi r^2 h$

$162 = 50h$

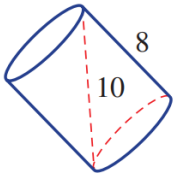
$h = 3.24$ cm

d Calculate the surface area of the cylinder.

171 cm²

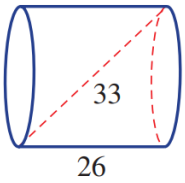
14 Use Pythagoras' Theorem $h^2 = a^2 + b^2$ to find unknown lengths and hence calculate the surface area of these solids. All units are in cm. Round your answer to 1 d.p.

a



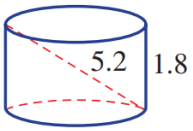
$r = 3$ cm
$SA = 207.4$ cm

b



$r = 10.1612\ldots$
$SA = 2308.7$ cm²

c



$r = 2.43926\ldots$
$SA = 65.0$cm²

15 Tennis balls are tightly packed in cylindrical cans which hold three balls.

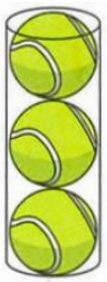
The radius of a ball is 37 mm.

The curved surface is manufactured from thin aluminium which costs \$4.80/m².

The top and bottom are made from stronger aluminium which costs \$5.60/m².

At the Australian Open Tennis Championships, about 48 000 balls are used.

Find the cost of producing the cans to hold this number of balls.



Dimensions of can: $r = 37$ mm, $h = 222$ mm

Circular ends = $2\pi(37)^2 = 2738\pi$ mm²

Curved surface = $2\pi(37)(222) = 16428\pi$ mm²

Cans needed = $48000 \div 3 = 16\,000$

Strong aluminium needed = $2738\pi \times 16000 = 43\,808\,000\pi$ mm² ÷ 1000² = 43.808π m²

Thin aluminium needed = $16428\pi \times 16000 = 262\,848\,000\pi$ mm² ÷ 1000² = 262.848π m²

Cost of thin aluminium = $43.808\pi \times \$5.60 = \$770.7105894\ldots$

Cost of strong aluminium = $262.848\pi \times \$4.80 = \$3963.65446\ldots$

Total cost = \$4734.37

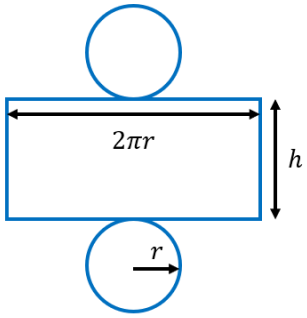
Check your answer using a faster method: Finding the cost of 1 can, then multiplying by 16 000.

Surface Area of Cylindrical Solids

ⓘ Surface Area of a Closed Cylinder

$$SA = 2\pi r^2 + 2\pi rh$$

2 circles — curved surface

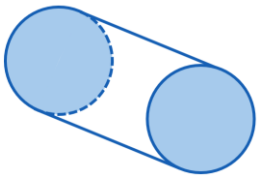


- Many questions will ask you to find a **part** of a cylinder.
- You need to decide how to adjust the formula:
  - Is there a full circular base, part of a base, or no base?
  - How much of the curved surface area remains?
- Is there a new flat surface where the cylinder was "cut"?

🔍 **Identify** Formula for a part of a cylinder

Write an expression for the surface area of these solids; the radius is r and height is h.

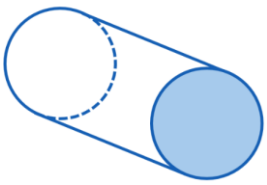
a Closed cylinder



Two end circles:

Curved surface:

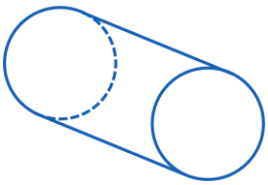
b Cylinder with one open end



One end circle:

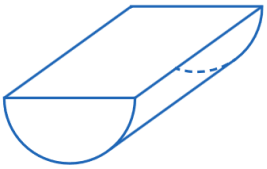
Curved surface:

c Cylinder open both ends



Curved surface:

d Closed half cylinder

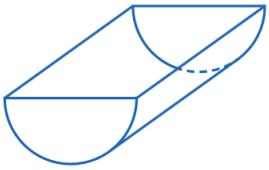


Two end semicircles:

Curved surface:

Top rectangle:

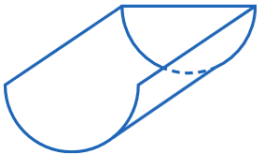
e Half cylinder with open top



Two end semicircles:

Curved surface:

f Half cylinder with open top, one end open

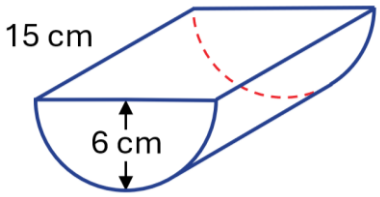


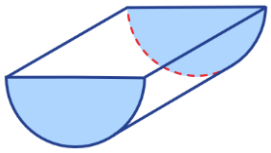
One end semicircle:

Curved surface:

Example Calculate surface area of cylindrical solids

Calculate the surface area of this **closed** half cylinder, rounded to 2 decimal places.

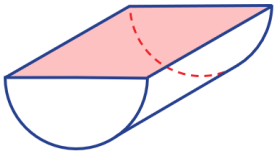




front and back semicircles

$\left(\frac{1}{2} \times \pi(6)^2\right) \times 2$

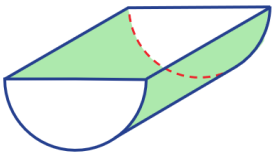
$= 113.0973 \ldots$



top rectangle

15×12

$= 180$



curved surface (rectangle)

$\frac{1}{2} \times 2\pi(6)(15)$

$= 282.7433 \ldots$

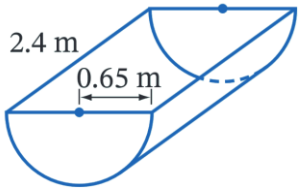
Total surface area $= 113.0972 + 180 + 282.7433$

$= 575.84 \text{ cm}^2$

Guided Practice

Calculate the surface area of these solids.
Give intermediate answers to 4 d.p. and final answer to 2 d.p.

a Half-cylinder, **open top**



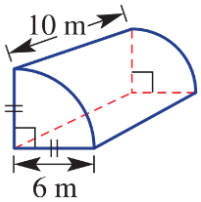
Front and back:

Curved side:

Total surface area:

6.23 m²

b **Closed** quarter-cylinder



Front and back:

Curved side:

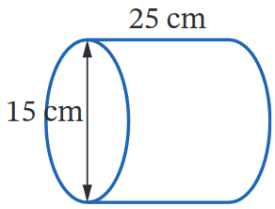
Side rectangles:

Total surface area:

270.80 m²

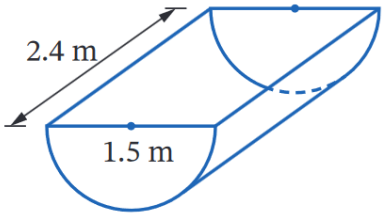
1 Calculate the surface area of the following solids to the nearest whole number.

a Cylinder with one open end



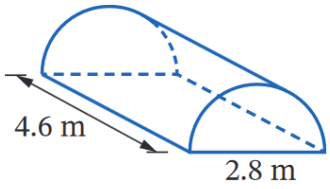
1355 cm²

b Half cylinder with open top



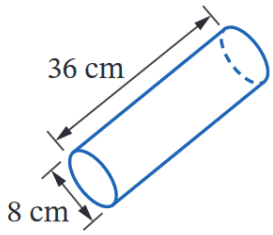
7 cm²

c Closed half-cylinder



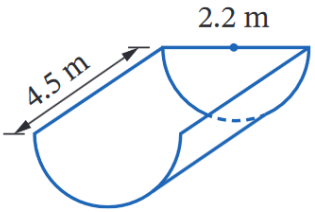
39 m²

d Cylinder open both ends



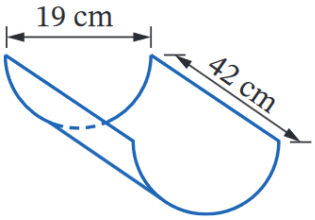
905 cm²

e Half cylinder with open top, one end open



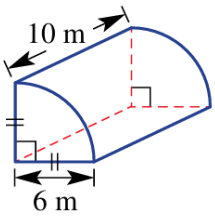
17 m²

f Half-cylinder with open top, open both ends



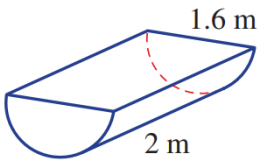
1253 cm²

g Closed quarter-cylinder



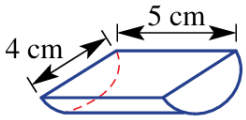
271 m²

h Half-cylinder with open top



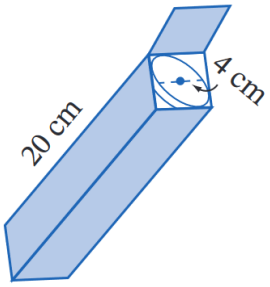
7 m²

i Closed half-cylinder



64 cm²

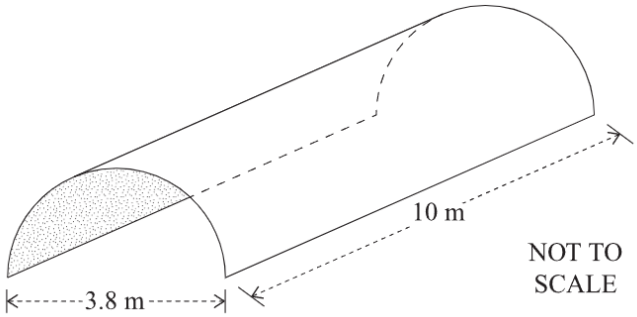
2 Circular cracker biscuits of diameter 4 cm are packed in a cardboard box of length 20 cm.
How much cardboard would be saved if the biscuits were packed into a cylindrical box instead?



76 cm²

3 **2018 HSC Standard 2 Band 4**

A shade shelter is to be constructed in the shape of half a cylinder with open ends.
The diameter is 3.8 m and the length is 10 m.
The curved roof is to be made of plastic sheeting.
What area of plastic sheeting is required, to the nearest m²?

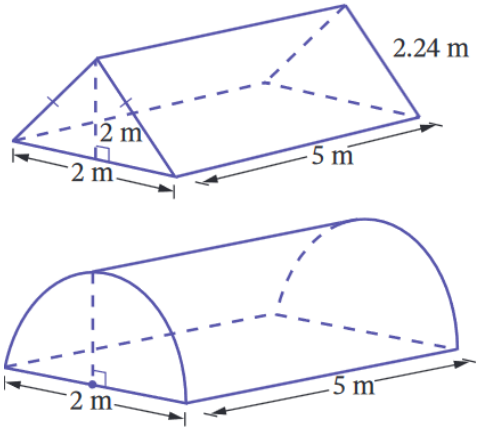


$A = \frac{1}{2}(2\pi rh)$

$= 2\pi(1.9)(10)$

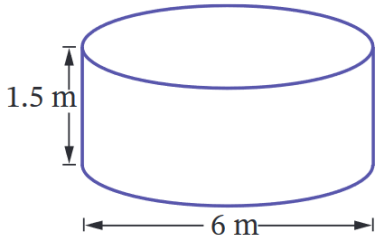
$= 60$ m²

4 Which tent has the greater surface area? (include the floors).



The triangular prism tent has 7.6 m² more surface area.

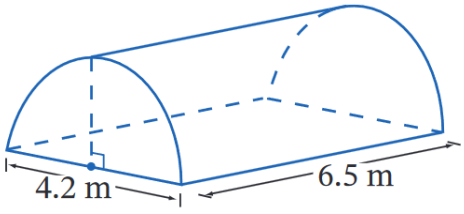
5 Jessica is repainting a swimming pool in the shape of a cylinder 1.5 m tall and 6 m in diameter.
The inside of the pool is to be repainted, including the floor.
If a one-litre can of paint covers 9 m^2, find the number of cans needed.



A = 56.5 m

6.3 L of paint required, so 7 cans to paint the pool.

6 The diagram shows a tent to be made in the shape of a half-cylinder.
The manufacturer knows that the flooring costs \$18.50/m^2 and the top canvas costs \$21/m^2.
How much would each tent cost to make? Answer to the nearest dollar.



Floor: 27.3 m²

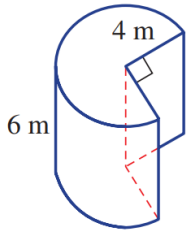
Surface Area: 56.7 m²

Cost = 27.3 × \$18.50 + 56.7 × \$21.75 = \$1738

7 Find the surface area of these closed partial cylinders. Answer in exact form.

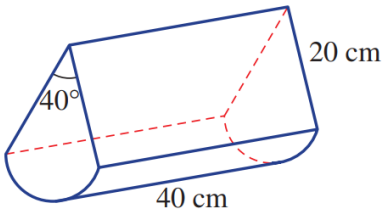
(hint: arc-length is $l = \frac{\theta}{360} \times 2\pi r$)

a



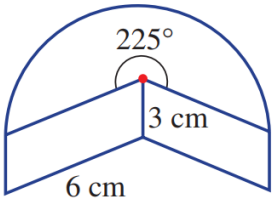
$(60\pi + 48)$ m²

b



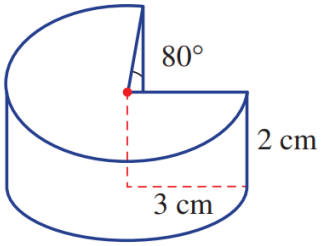
$\left(\frac{800\pi}{3} + 1600\right)$ cm²

c



$\left(\frac{135\pi}{2} + 36\right)$ cm²

d

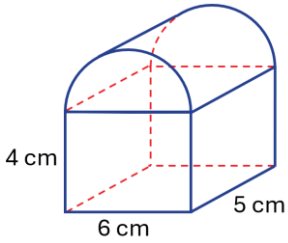


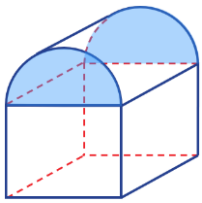
$\left(\frac{70\pi}{3} + 12\right)$ cm²

Surface Area of Composite Prisms

Example Calculate surface area of composite solids

Calculate the surface area of this solid, rounded to 2 decimal places.

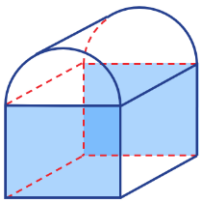




base semicircles

$\left(\frac{1}{2} \times \pi(3)^2\right) \times 2$

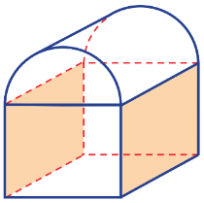
$= 28.2743 \ldots$



base rectangles

$(4 \times 6) \times 2$

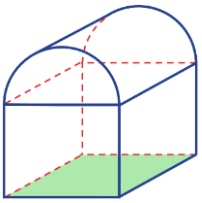
$= 48$



side rectangles

$(4 \times 5) \times 2$

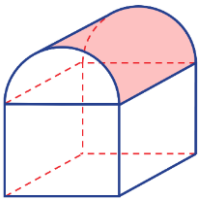
$= 40$



bottom rectangle

6×5

$= 30$



top surface (rectangle)

$\frac{1}{2} \times 2\pi(3)(5)$

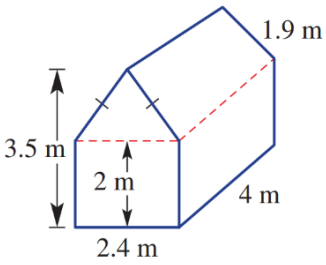
$= 47.1238 \ldots$

Total surface area $= 28.2743 + 48 + 40 + 30 + 47.1238$

$= 193.40 \text{ cm}^2$

Guided Practice

Calculate the surface area of this composite solid.
Give final answer to the nearest whole.



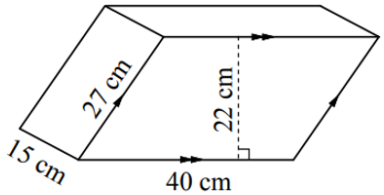
base triangles | base rectangles | side rectangles | top rectangles | bottom rectangle

Total surface area =

54 m²

1 Find the surface area of these solids. Round to 2 d.p.

a



SA = 2 × □ × □ Area of front and back

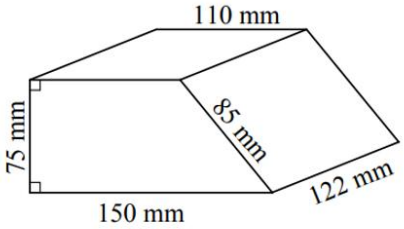
+ 2 × □ × □ Area of top and base

+ 2 × □ × □ Area of left and right sides

SA =

3770 cm²

b



SA = □ × □ + □ × □

Area of top Area of base

+ □ × □ + □ × □

Area of LHS face Area of slanted face

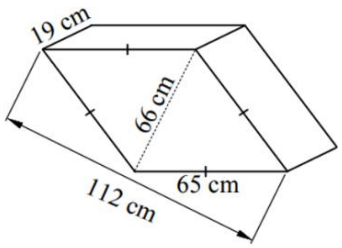
+ 2 × ½ × (□ + □) × □

Area of the front and back faces (trapezia)

SA =

70 740 mm²

c



SA = 2 × ½ × □ × □

Front and Rear faces (Rhombi)

+ 4 × □ × □

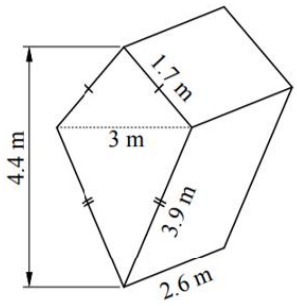
Four sides

SA =

12 332 cm²

2 Find the surface area of these solids.
Set your working out neatly to show that you have considered every face of the solid.

e



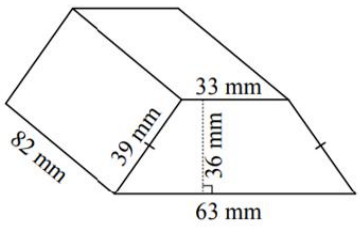
Front & Back: $\left(\frac{3 \times 4.4}{2}\right) \times 2$

Top two: $(1.7 \times 2.6) \times 2$

Bottom two: $(3.9 \times 2.6) \times 2$

Total: 42.32 m²

f



Front & Back: $\left(\frac{36}{2}(33 + 63)\right) \times 2$

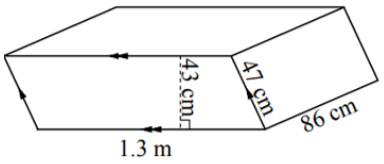
Left & Right: $(39 \times 82) \times 2$

Top: 33×82

Bottom: 63×82

Total: 17 724 mm²

g Answer in m²



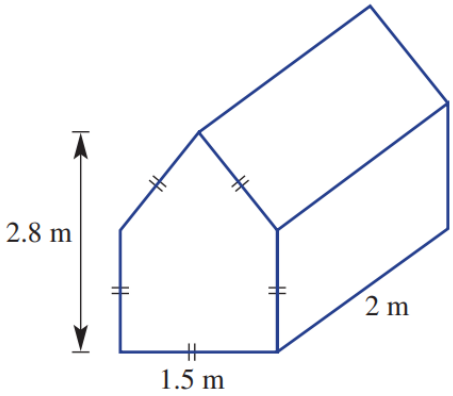
Front & Back: $(1.3 \times 0.43) \times 2$

Left & Right: $(1.3 \times 0.86) \times 2$

Top & Bottom: $(0.47 \times 0.86) \times 2$

Total: 4.1624 m²

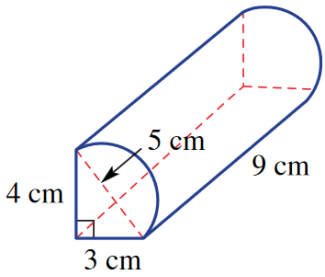
3 The floor, sides and roof of this tent are made from canvas at a cost of \$5 per square metre. The tent's dimensions are shown in the diagram. What is the cost of the canvas for the tent?



Surface area = 21.45 m²

Cost: \$42.90

4 Find the surface area of this composite prism to 2 d.p.



Front & back: $2\left(\frac{4\times3}{2}+\frac{\pi(2.5)^2}{2}\right)$

Left: 4×9

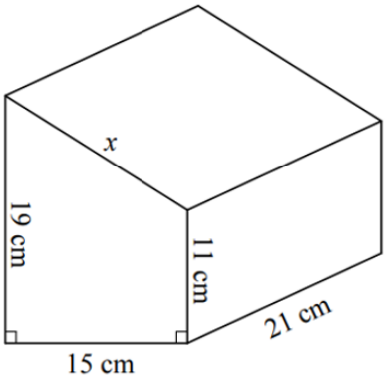
Bottom: 3×9

Right: $\frac{2\pi(2.5)(9)}{2}$

Total: 165.32 cm²

5 Find the surface area of these solids.
First calculate the missing length using Pythagoras' Theorem.

a



$x = 17$ cm

Front & Back $\frac{15}{2}(19+11)\times 2$

Left: 15×19

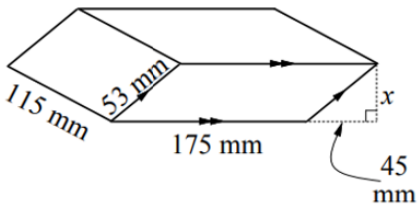
Right: 11×21

Top: 17×21

Bottom: 15×21

$SA = 1752$ cm²

b



$x = 28$ mm

Front & Back $(175 \times 28) \times 2$

Left & Right: $(115 \times 53) \times 2$

Top & Bottom: $(115 \times 175) \times 2$

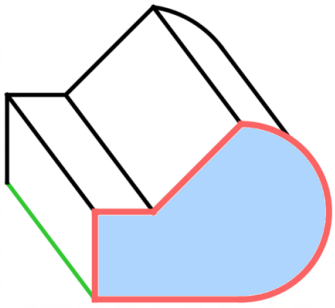
$SA = 62\,240$ mm²

6 The formula for calculating any solid with *uniform cross-section* is:

Surface Area of any Solid with Uniform Cross-Section

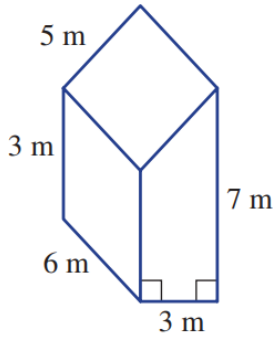
$$SA = 2B + ph$$

B: base area; p: base perimeter; h: height of prism



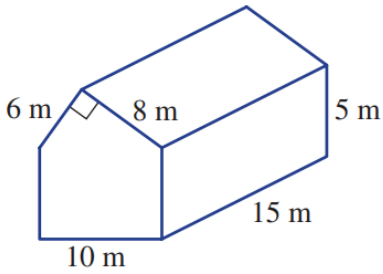
Calculate the surface area of these solids using the formula. Answer to 1 d.p. where needed.

a



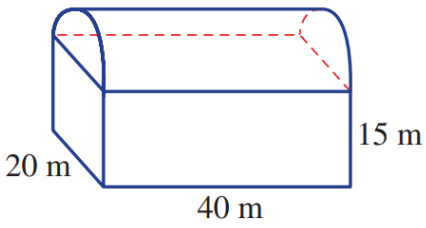
138 m²

b



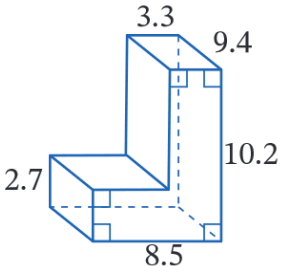
658 m²

c



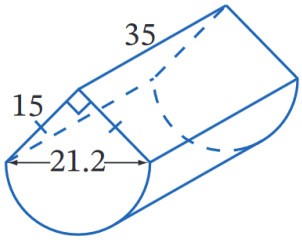
4170.8 m²

d



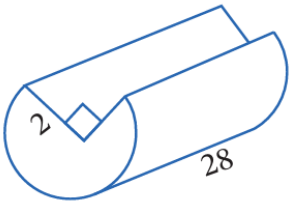
447.0 cm²

e



2793.5 cm²

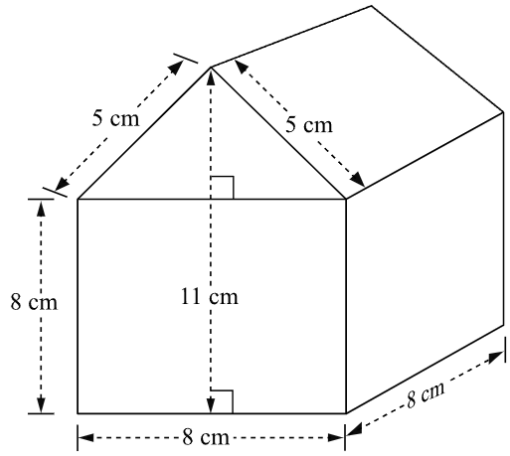
f



394.7 cm²

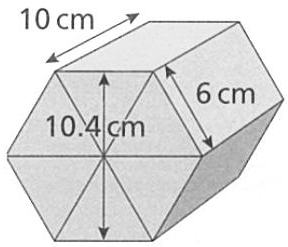
7 **2020 HSC Standard 2 Band 4**

Calculate the surface area of the composite solid.



424 cm²

8 Amina cuts a regular hexagonal prism into six, identical triangular prisms.
Work out the surface area of each triangular prism rounded to 1 decimal place.



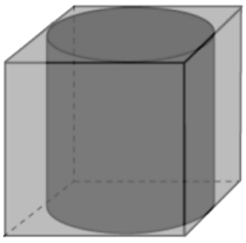
The triangular faces are equilateral.

Area of triangles: $\frac{5.2\times6}{2}\times2$

Area of rectangular faces: $(6\times10)\times3$

Total SA: 211.2 cm²

9 A cube with surface area 216 square centimetres has a cylinder perfectly enclosed inside.
What is the surface area of the cylinder? Answer correct to 2 decimal places.



Find cube side length: $6x^2 = 216$

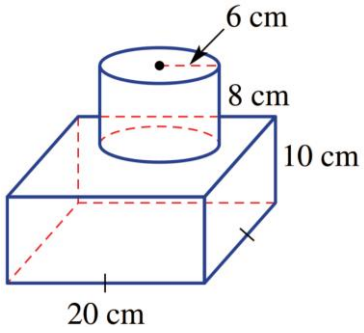
$x = 6$ cm

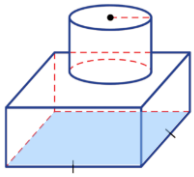
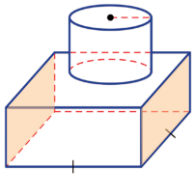
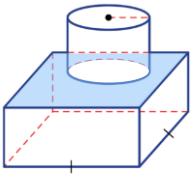
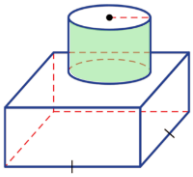
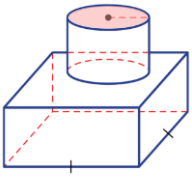
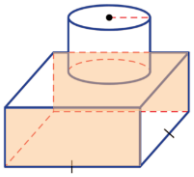
SA of cylinder $= 2\pi(3)^2 + 2\pi(3)(6) = 169.65$ cm²

Surface Area of Prism Combinations

Example Calculate surface area of composite solids

Calculate the surface area of this solid, rounded to 1 decimal place.



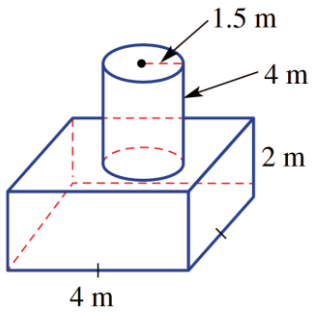







bottom square	side rectangles	top square – circle	curved surface (rectangle)	top circle
20×20	$(10 \times 20) \times 4$	$20 \times 20 - \pi(6)^2$	$2\pi(6)(8)$	$\pi(6)^2$
$= 400$	$= 800$	$= 286.90266 \ldots$	$= 301.59228 \ldots$	$= 113.09733 \ldots$

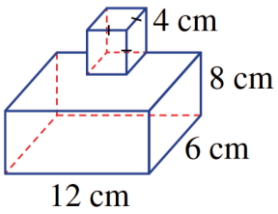
Total surface area $= 1901.6\ \text{cm}2$

Guided Practice

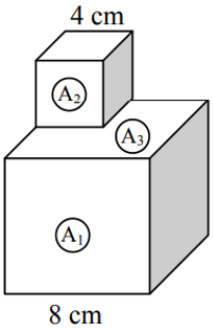
a Show that the surface area of this composite prism is 101.7 m^2.



b Show that the surface area of this composite prism is 496 m^2.
Note that the base is not a square.



1 Find the surface area of the cube.



Cube with sides 4 cm length sits on top of a cube with sides 8 cm.

$A_1 = \square \times \square =$

$A_2 = \square \times \square =$

$A_3 = = A_1 - A_2$

$= \square - \square =$

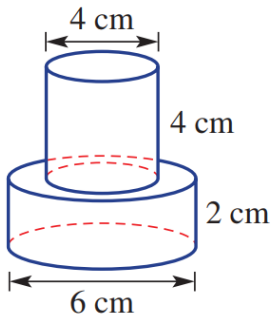
$SA = 5 \times \square + 5 \times \square + \square$

$SA =$

448 cm²

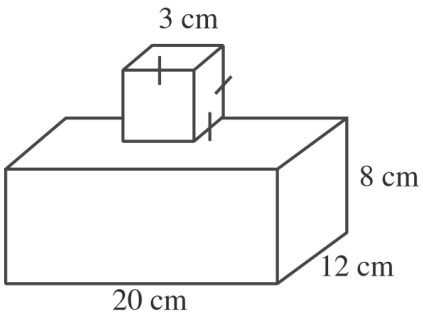
2 Find the surface area of these composite solids. Round to 1 d.p. where appropriate.

a



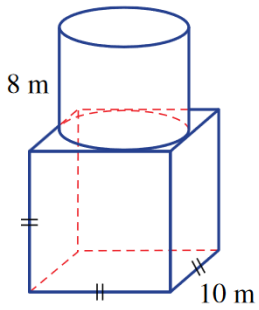
144.5 cm²

b



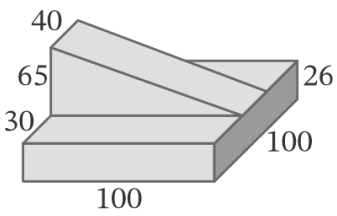
1028 cm²

c



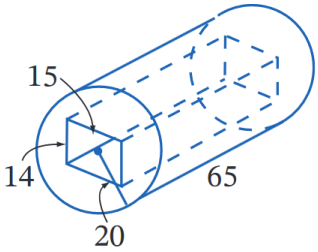
851.3 m²

d All measurements in cm.



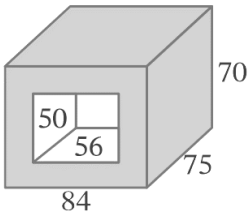
40 270 cm²

e All measurements in cm.



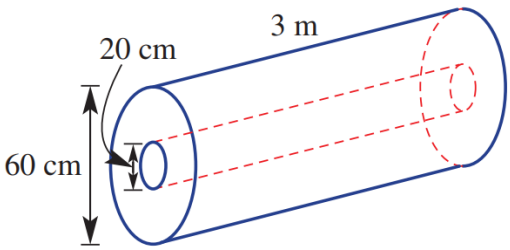
14 031.4 cm²

f All measurements in cm.



45 160 cm²

3 A log with diameter 60 cm is 3 m in length. Its hollow centre is 20 cm in diameter. Find the surface area of the log in cm², including the ends and the inside, correct to 1 decimal place.



Circular ends: $[\pi(30^2) - \pi(10^2)] \times 2$

Outside: $2\pi(30)(300)$

Inside: $2\pi(10)(300)$

Total: 80 424.8 cm²